CENTURY CLOCK

Nhóm 4

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Tống Tâm Ngọc

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1. Top module

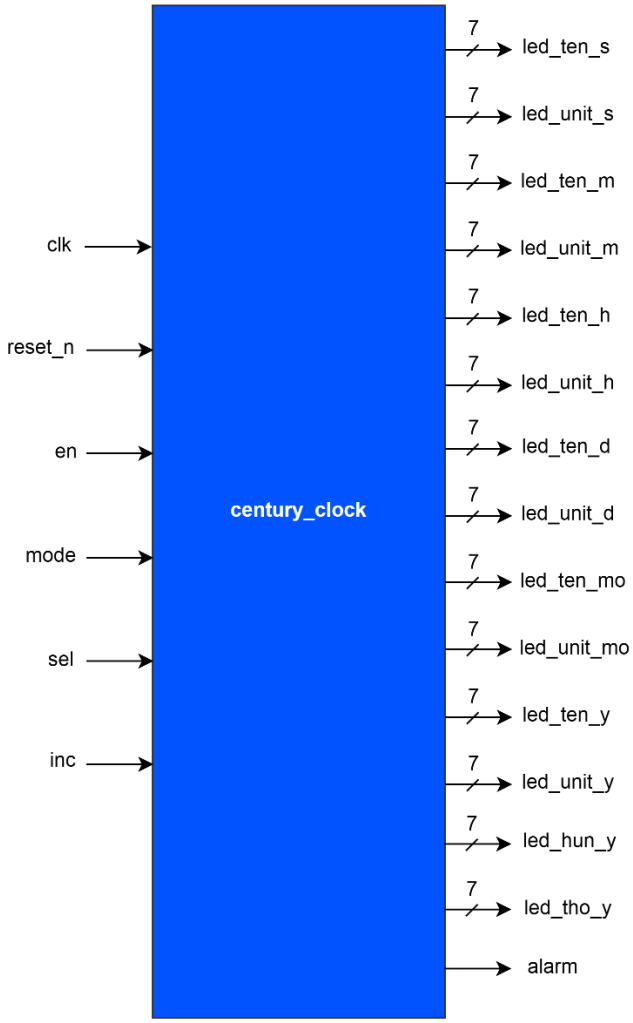

century_clock
clk
reset_n
en
mode
sel
inc
7
led_ten_s
7
led_unit_s
7
led_ten_m
7
led_unit_m
7
led_ten_h
7
led_unit_h
7
led_ten_d
7
led_unit_d
7
led_ten_mo
7
led_unit_mo
7
led_ten_y
7
led_unit_y
7
led_hun_y
7
led_tho_y
alarm

Figure 1.Century block diagram

1.1. Interface signal

Signal Name	Width	I/O	Description
clk	1	Input	System clock signal
reset_n	1	Input	System asynchronous reset, active LOW
en	1	Input	Signal to enable clock century
mode	1	Input	Signal to choose mode for clock
sel	1	Input	Signal to choose mode for clock
inc	1	Input	Signal to increase time
led_ten_s	7	Output	Seven seg second tens
led_unit_s	7	Output	Seven seg second units
led_ten_m	7	Output	Seven seg minute tens
led_unit_m	7	Output	Seven seg minute units
led_ten_h	7	Output	Seven seg hour tens
led_unit_h	7	Output	Seven seg hour units
led_unit_y	7	Output	Seven seg year units
led_ten_y	7	Output	Seven seg year tens
led_hun_y	7	Output	Seven seg yearhunds
led_thou_y	7	Output	Seven seg year thous

Table 1. century block signals description

1.2. Functional Description:

Description:

- Clock Century is a simple digital clock that performs the following main functions:
- Count time: It counts seconds, minutes, and hours in real time.
- Display time: It shows the current second, minute, and hour on seven-segment LEDs.
- Reset automatically: When seconds reach 60, it increases the minute by one, and when minutes reach 60, it increases the hour by one.
- Simple design: It is easy to understand and can be used for learning basic digital clock operation.

1.3. Architecture

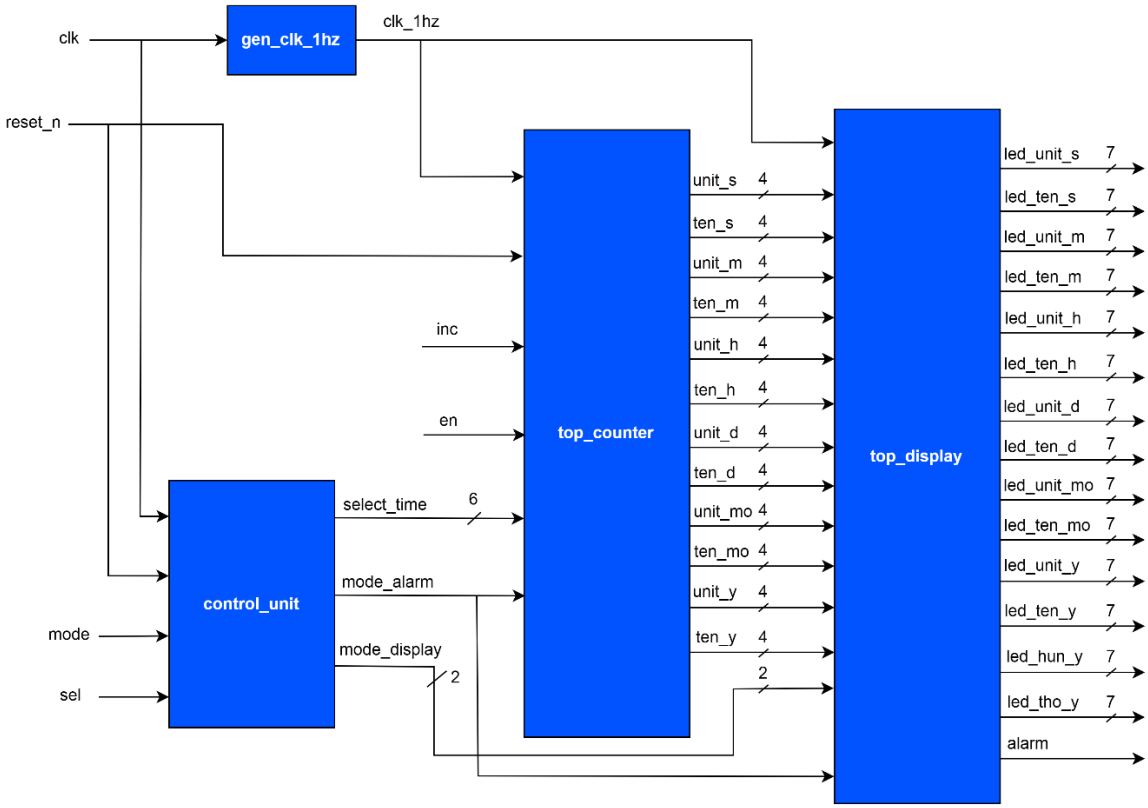


Figure 2. century clock architecture

Figure 2 shows the architecture of clock century. The main functions are performed in **count_second, count_min** and **count_hour**.

2. Module gate level

2.1. mux 2_1



Figure 3. mux2_1 block diagram

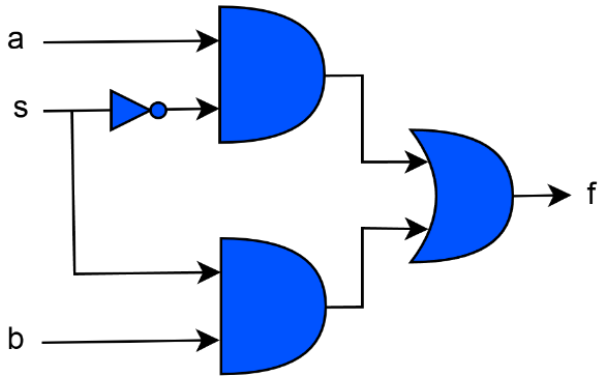


Figure 4. mux2_1 block diagram

2.1.1. Signal

Signal Name	Width	I/O	Description
s	1	Input	Signal to choose data
a	1	Input	Data in
b	1	Input	Data in
f	1	Output	Data out

Table 2. mux2_1 signals description

2.2. half_adder



Figure 5. half_adder block diagram

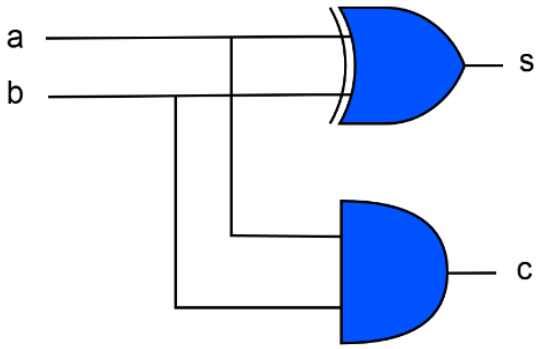


Figure 6. half_adder block diagram

2.2.1. Signal

Signal Name	Width	I/O	Description
a	1	Input	Bit 1
b	1	Input	Bit 2
s	1	Output	Sum
c	1	Output	Carry out

Table 3. half_adder signals description

2.3. full_adder



Figure 7. full_adder block diagram

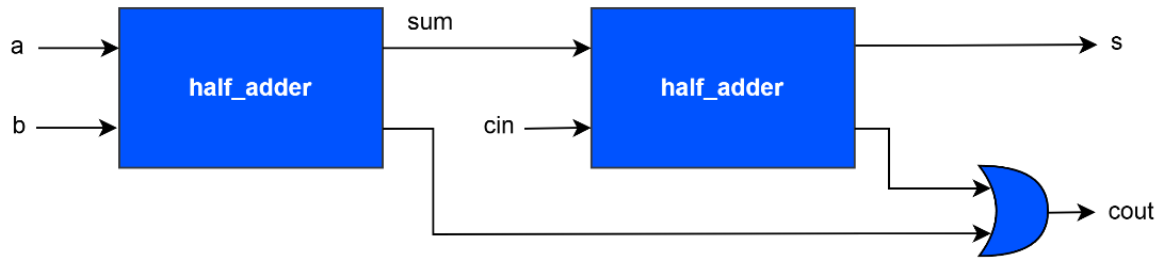


Figure 8. full_adder block diagram

2.3.1. Signal

Signal Name	Width	I/O	Description
a	1	Input	Bit 1
b	1	Input	Bit 2
cin	1	Input	Carry in
s	1	Output	Sum
cout	1	Output	Carry out

Table 4. full_adder signals description

2.4. adder_4bit

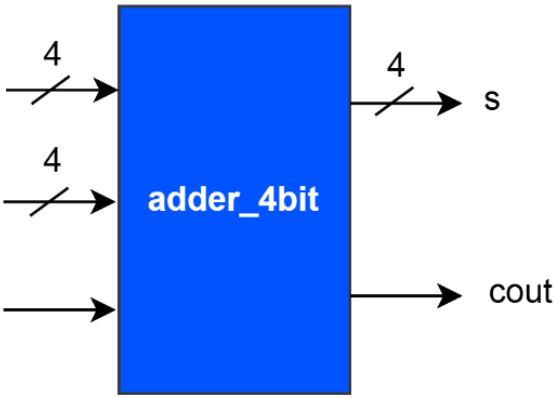


Figure 9. adder_4bit block diagram

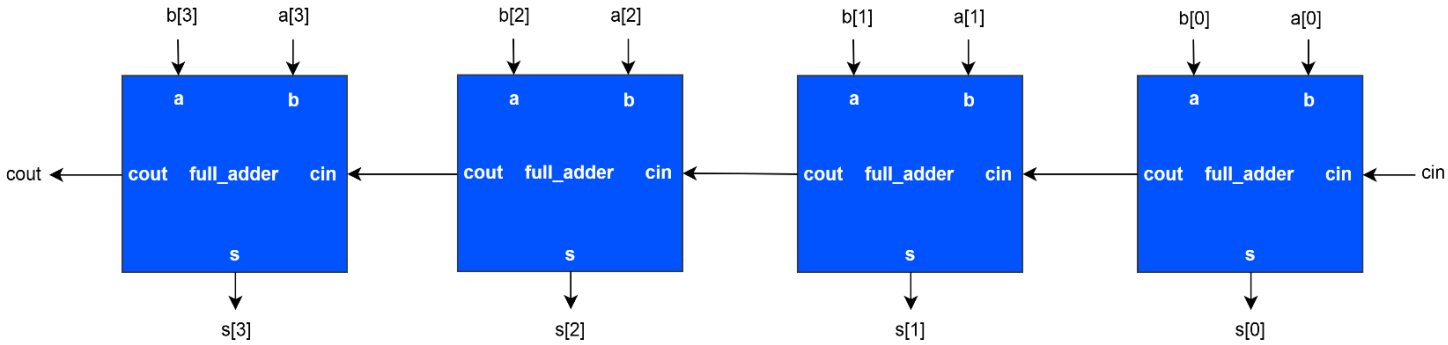


Figure 10. adder_4bit block diagram

2.4.1. Signal

Signal Name	Width	I/O	Description
a	4	Input	Operand 1
b	4	Input	Operand 2
cin	1	Input	Carry in
s	4	Output	Sum
cout	1	Output	Carry out

Table 5. adder_4bit signals description

2.5. adder_4bit



Figure 11. equality_comparator_4bit block diagram

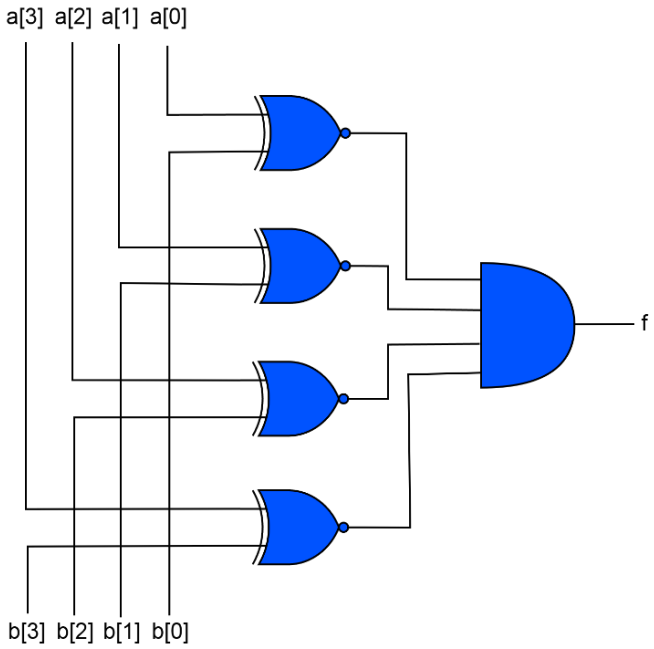


Figure 12. equality_comparator_4bit block diagram

2.5.1. Signal

Signal Name	Width	I/O	Description
a	4	Input	Operand 1
b	4	Input	Operand 2
f	1	Output	Result

Table 6. equality_comparator_4bit signals description

3. Sub module

3.1. control_unit

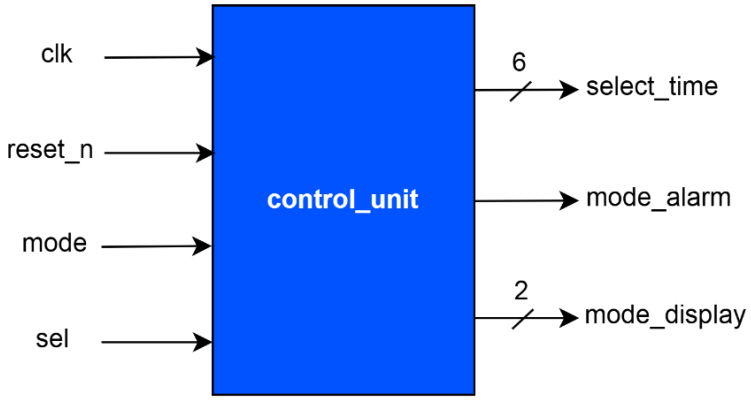


Figure 13. control_unit block diagram

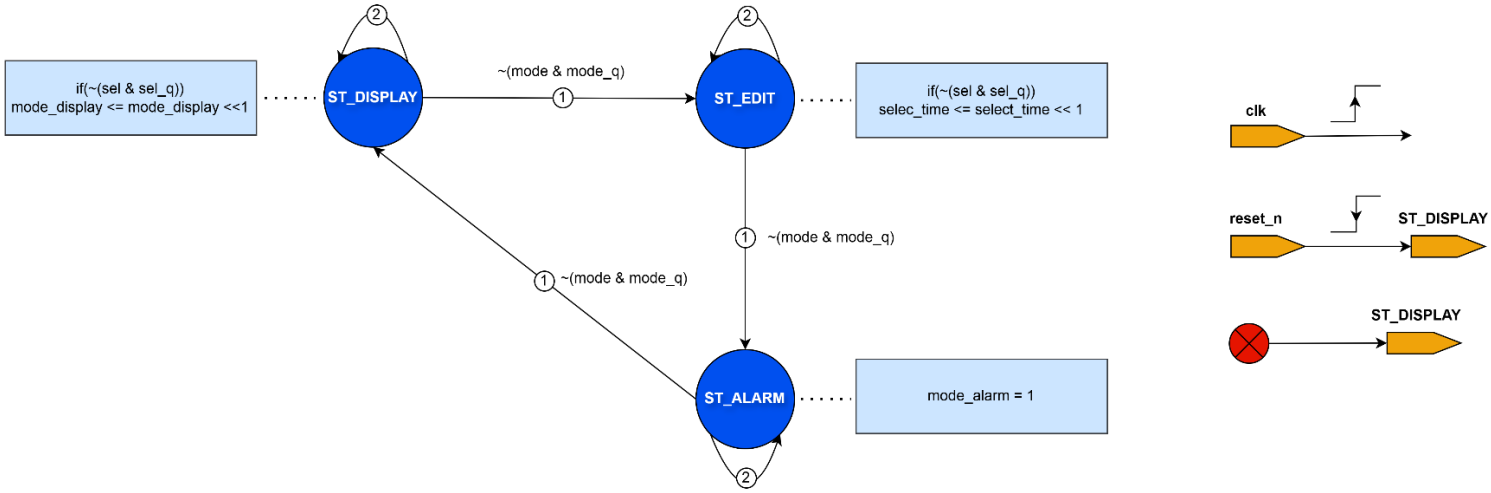


Figure 14. control_unit finite state machine

The control_unit finite state machine (FSM) manages the main operation modes of the clock system. It has three states: ST_DISPLAY, ST_EDIT, and ST_ALARM.

ST_DISPLAY:
This is the default state after reset. In this state, the current time is shown on the display. When the sel signal is activated, the display mode (mode_display) shifts to the next value.

ST_EDIT:
When the mode signal is active, the FSM switches to the edit mode. In this mode, users can adjust the time settings. If sel is triggered, the select_time value shifts to choose which part of the time (hour, minute, or second) to edit.

ST_ALARM:
When both mode and mode_q signals satisfy the alarm condition, the FSM moves to the alarm state. In this state, mode_alarm is set to 1 to enable the alarm function.

The FSM transitions between these states based on the mode and sel signals, ensuring smooth switching between display, edit, and alarm operations. This design helps the clock system respond quickly to user inputs while maintaining stable control of the display and timing functions.

3.1.1. Interface signal

Signal Name	Width	I/O	Description
clk	1	Input	System clock signal
reset_n	1	Input	System asynchronous reset, active LOW
mode	1	Input	Signal to choose mode for clock
sel	1	Input	Signal to choose mode for clock
select_time	6	Output	Signal to choose time to edit
mode_alarm	1	Output	Signal to indicate mode alarm
mode_display	2	Output	Signal to indicate mode display

Table 7. control_unit signals description

3.2. top_counter

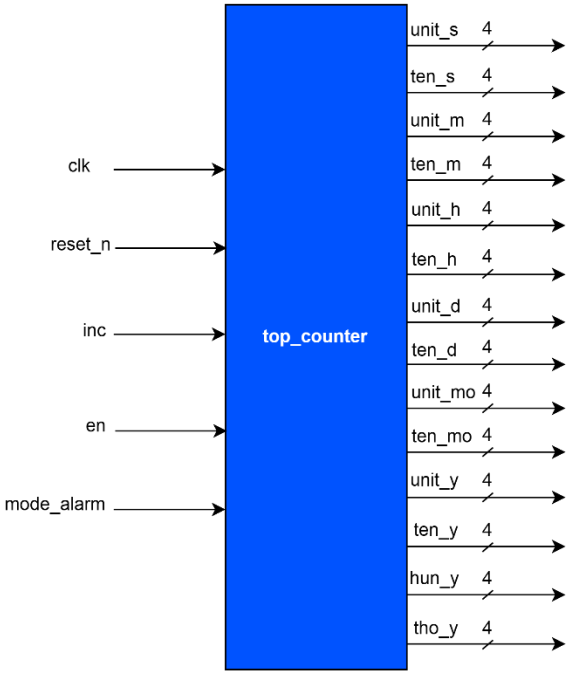


Figure 15. top_counter block diagram

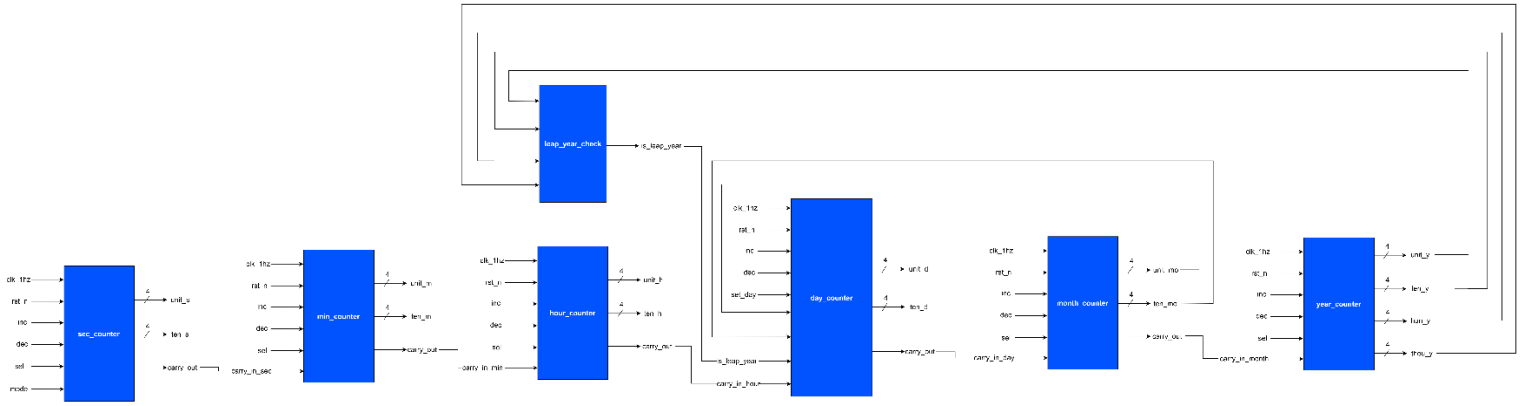


Figure 13. top_counter block diagram

3.2.1. second_counter

This block is responsible for counting seconds in the clock system. It receives an enable signal from the outside, which controls whether the counter is running or paused. When the enable signal is active, the block increases the second count step by step. If the signal is inactive, the counting process stops until it is enabled again.

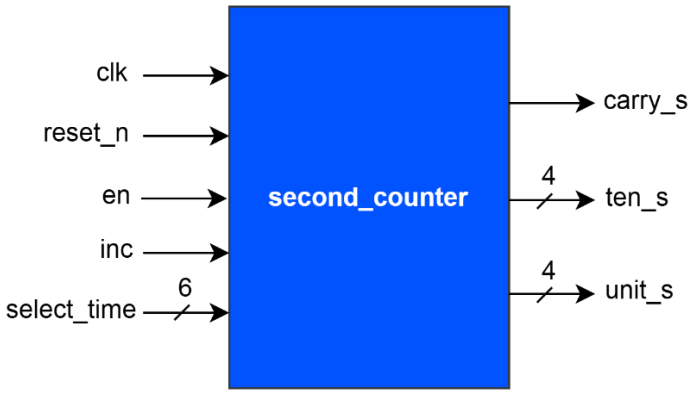


Figure 16. second_counter block diagram

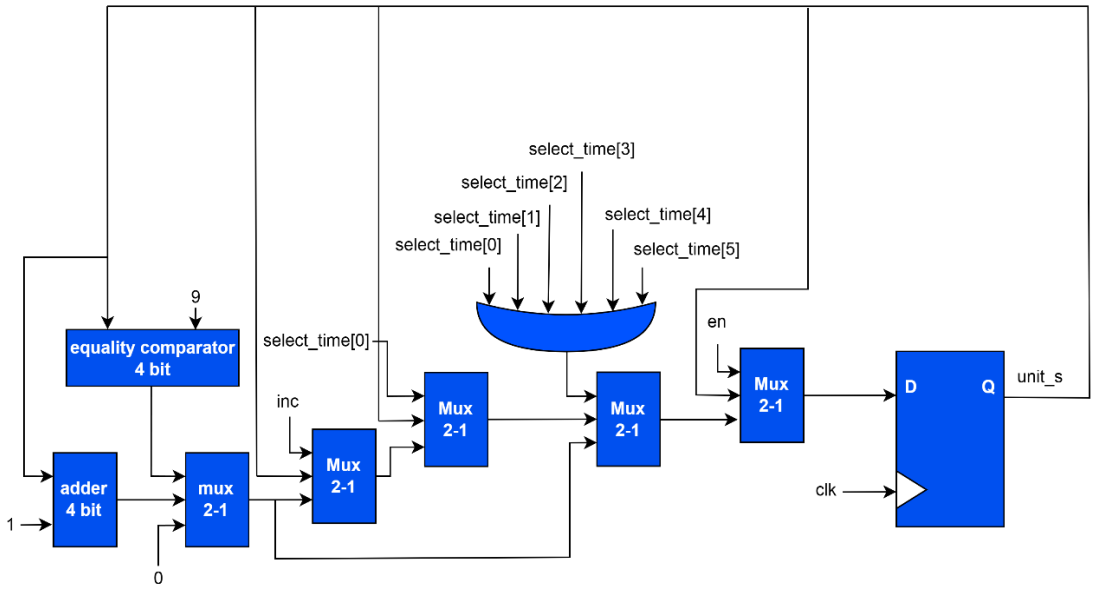


Figure 17. second_counter block diagram

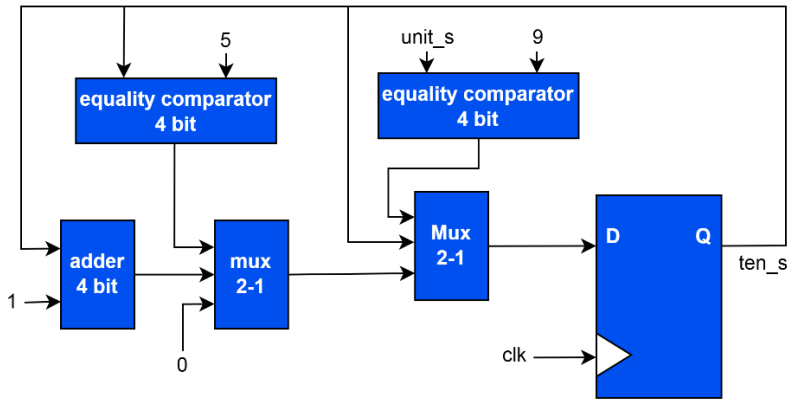


Figure 18. second_counter block diagram

3.2.1.1 Interface signal

Signal Name	Width	I/O	Description
clk	1	Input	System clock signal
reset_n	1	Input	System asynchronous reset, active LOW
en	1	Input	Signal to enable clock century
inc	1	Input	Signal to increase time
select_time	1	Input	Signal to chosse time to edit
carry_s	1	Output	Signal to indicate second is count to 59
ten_s	4	Output	Ten second counter
unit_s	4	Output	Unit second counter

Table 8. second_counter signals description

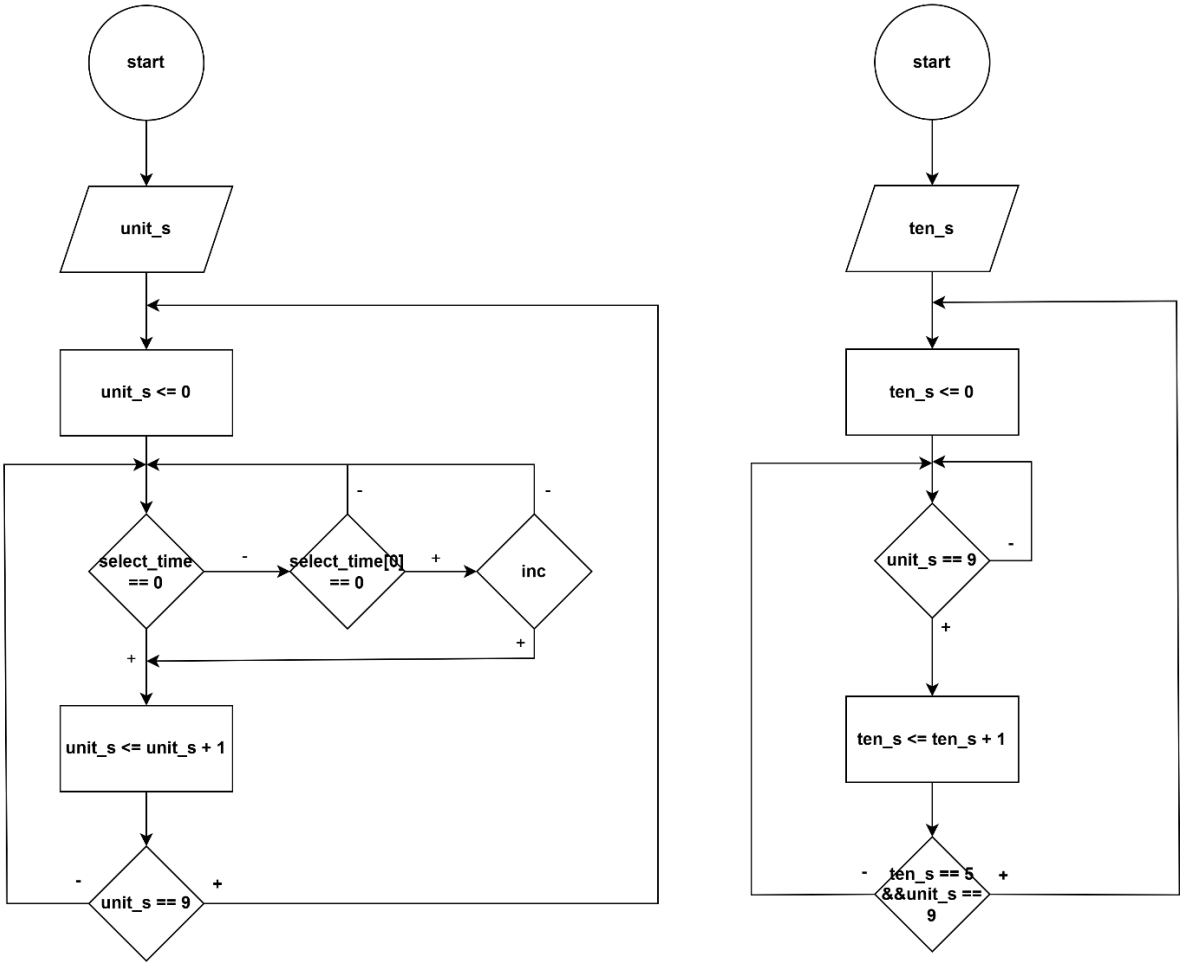


Figure 19. second_counter algorithm diagram

3.2.2. min_counter

This block is responsible for counting minutes and depends on the second counter for its operation. When the second counter reaches 60, this block increases the minute count by one. It continues to count up to 59 minutes and then resets to zero when an hour is completed.

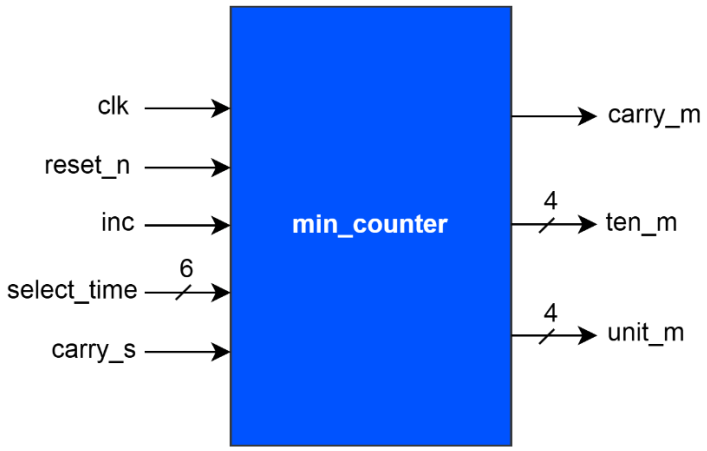


Figure 20. min_counter block diagram

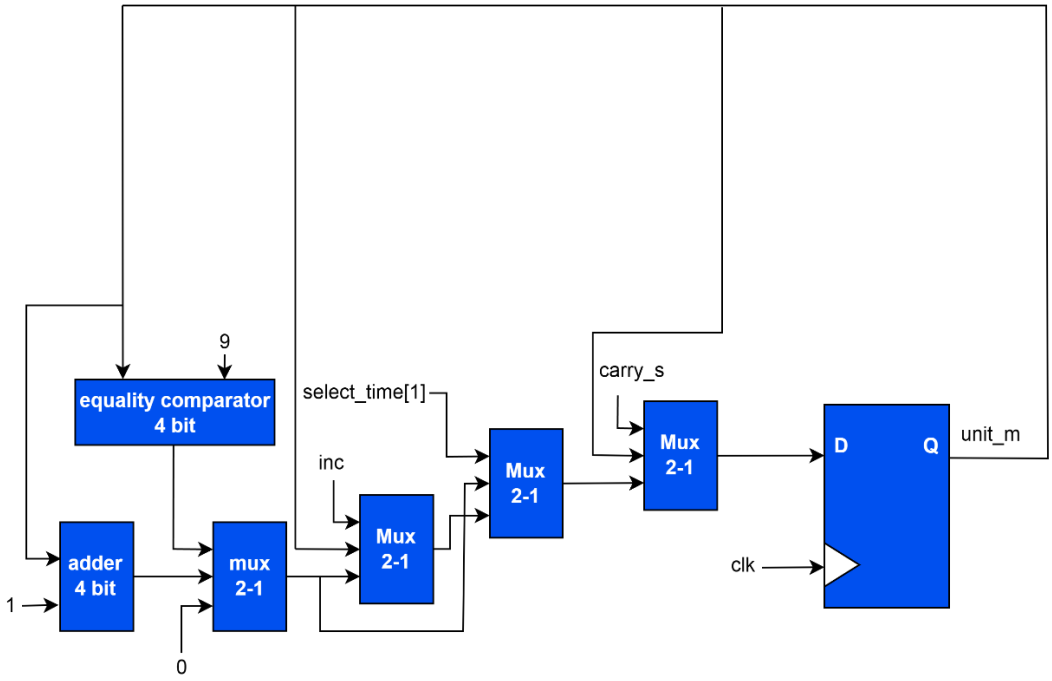


Figure 21. min_counter block diagram

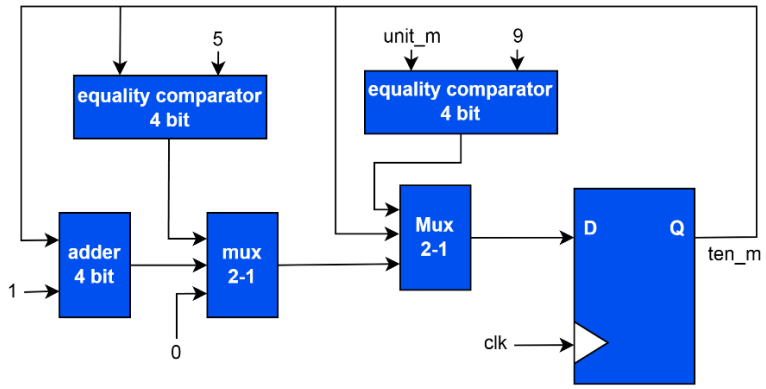


Figure 22. min_counter block diagram

3.2.1.1 Interface signal

Signal Name	Width	I/O	Description
clk	1	Input	System clock signal
reset_n	1	Input	System asynchronous reset, active LOW
carry_s	1	Input	Signal to indicate second is count to 59
inc	1	Input	Signal to increase time
select_time	1	Input	Signal to chosse time to edit
carry_m	1	Output	Signal to indicate minute is count to 59
ten_m	4	Output	Ten minute counter
unit_m	4	Output	Unit minute counter

Table 9. min_counter signals description

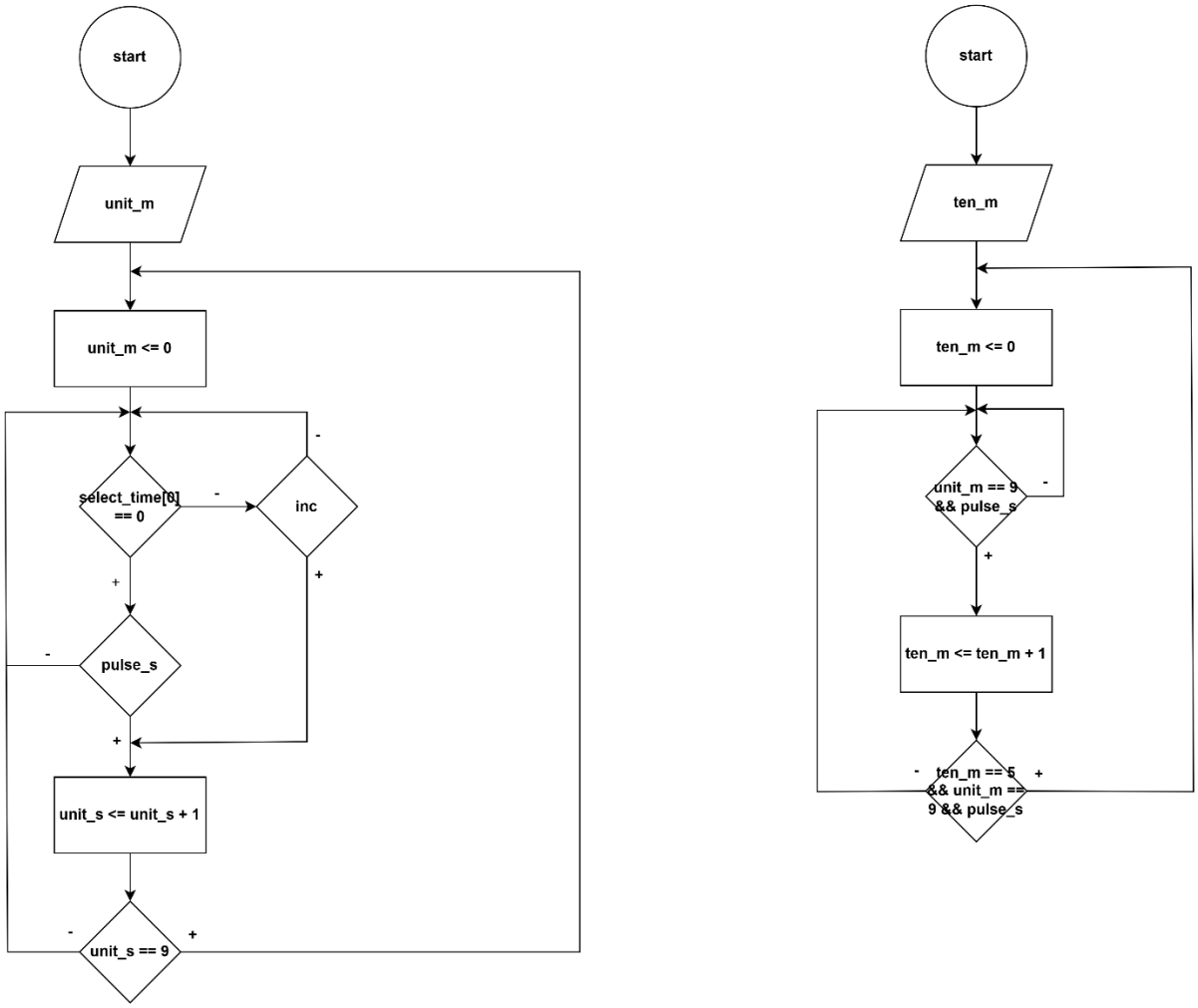


Figure 23. min_counter algorithm diagram

3.2.3. hour_counter

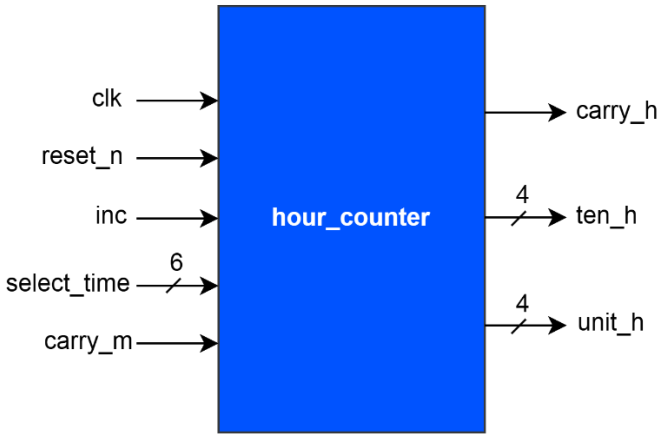


Figure 24. hour_counter block diagram

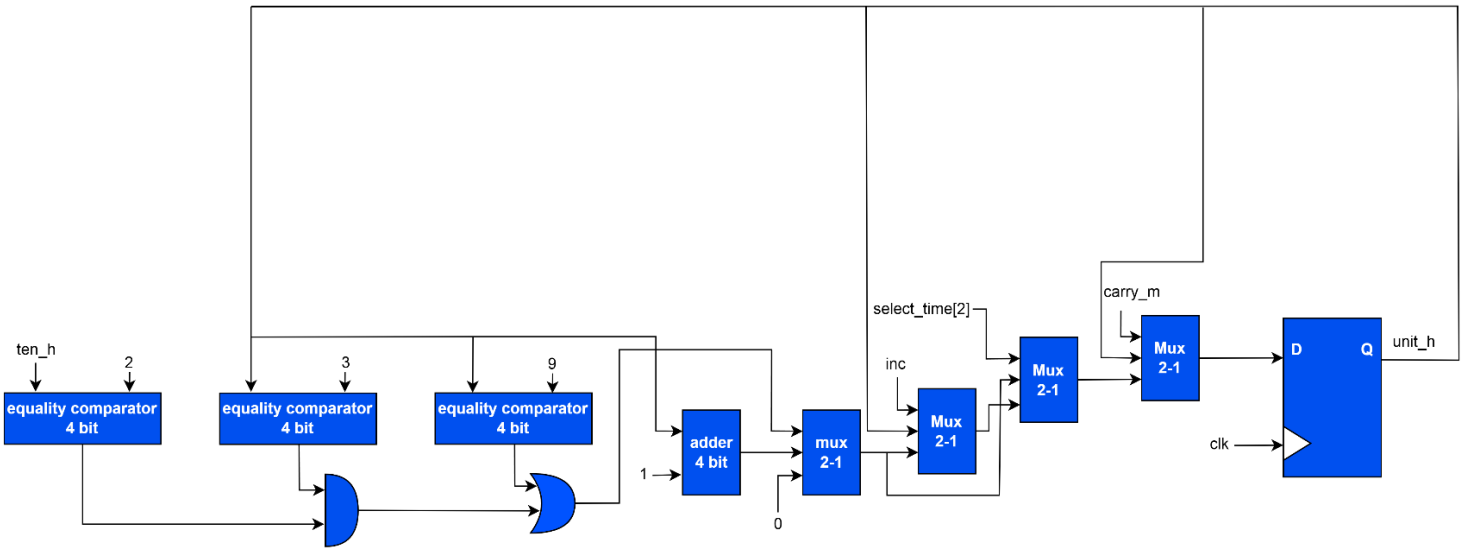


Figure 25. hour_counter block diagram

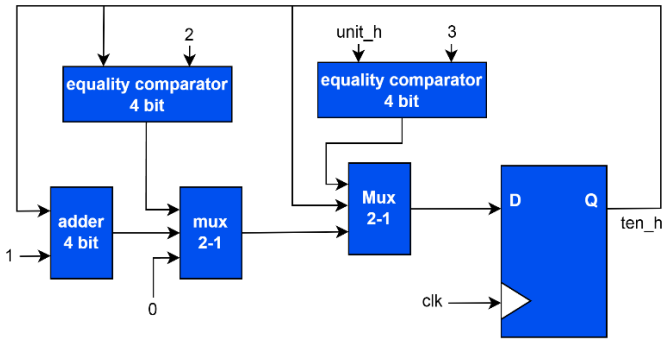


Figure 26. hour_counter block diagram

This block is responsible for counting hours and depends on both the second and minute counters. When the minute counter reaches 60, the hour counter increases by one. It continues counting up to 23 hours and then resets to zero to start a new day cycle.

3.2.3.1 Interface signal

Signal Name	Width	I/O	Description
clk	1	Input	System clock signal
reset_n	1	Input	System asynchronous reset, active LOW
carry_m	1	Input	Signal to indicate minute is count to 59
inc	1	Input	Signal to increase time
select_time	1	Input	Signal to chosse time to edit
ten_h	4	Output	Ten hour counter
unit_h	4	Output	Unit hour counter
carry_h	1	Output	Signal to indicate hour is count to 23

Table 9. hour_counter signals description
Table 10. hour_counter signals description

3.2.4. day_counter

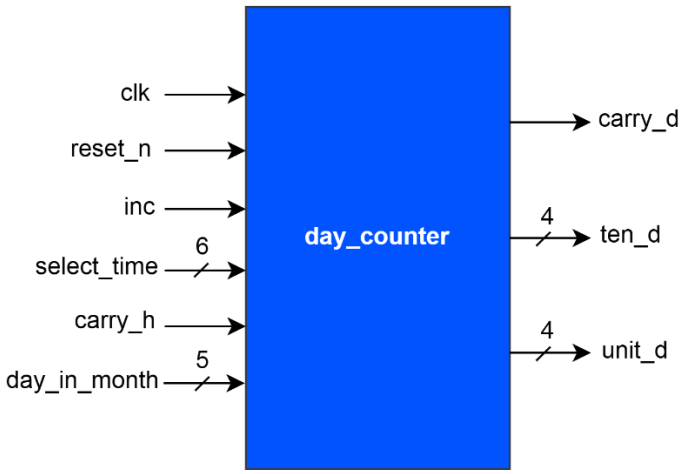


Figure 28. day_counter block diagram

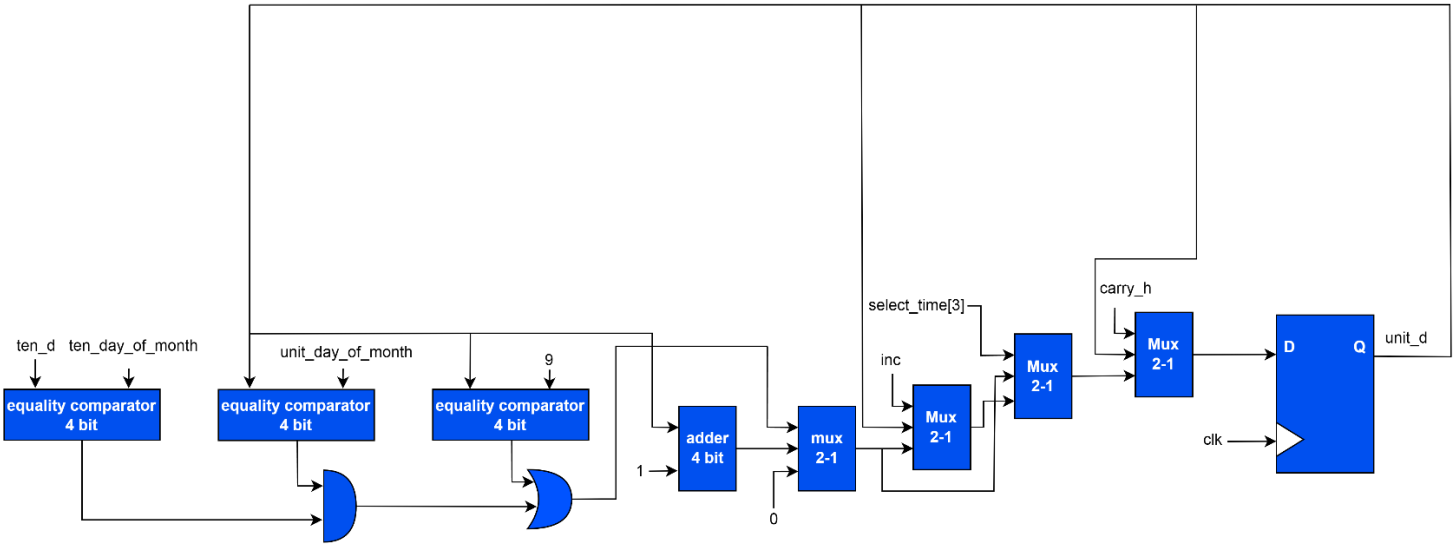


Figure 29. day_counter block diagram

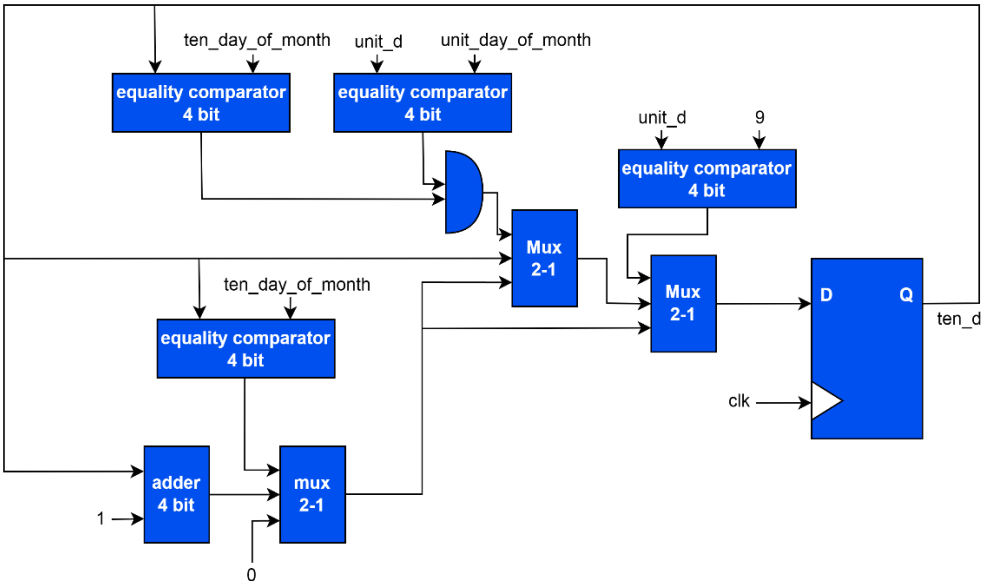


Figure 30. day_counter block diagram

This block is responsible for counting days and depends on the hour, seconds and minute counters. When the minute counter reaches 60, the hour counter increases by one. It continues counting up to 23 hours and then resets to zero to start a new day cycle.

3.2.4.1 Interface signal

Signal Name	Width	I/O	Description
clk	1	Input	System clock signal
reset_n	1	Input	System asynchronous reset, active LOW
carry_h	1	Input	Signal to indicate minute is hour to 23
inc	1	Input	Signal to increase time
select_time	1	Input	Signal to chosse time to edit
day_in_month	5	Input	Signal to indicate day in month
ten_d	4	Output	Ten day counter
unit_d	4	Output	Unit day counter
carry_d	1	Output	Signal to indicate day is count to day_in_month

Table 10. day_counter signals description

3.2.5. month_counter

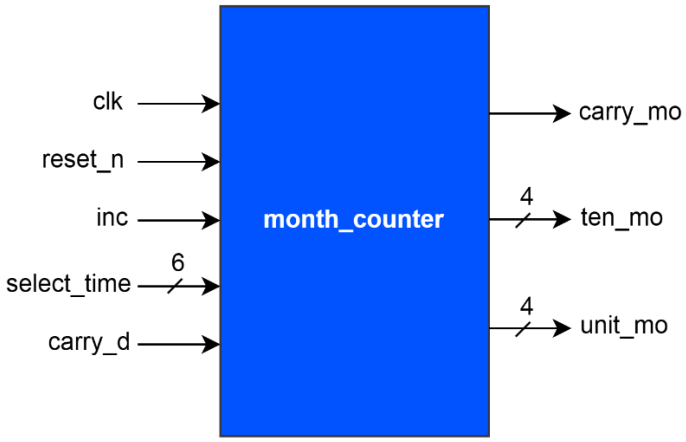


Figure 31. month_counter block diagram

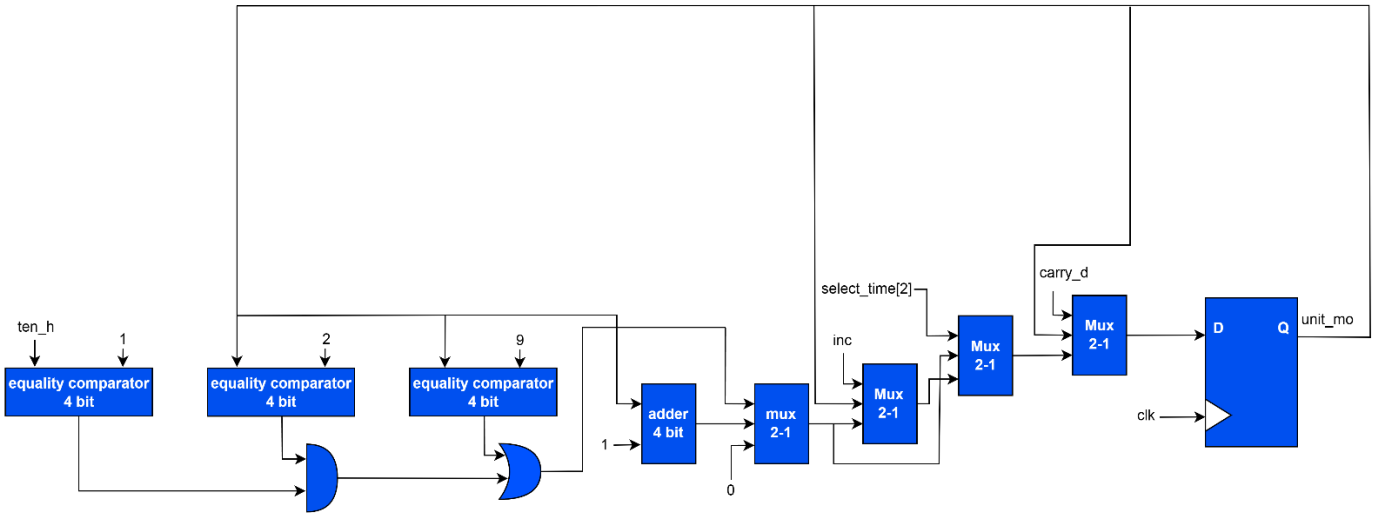


Figure 32. month_counter block diagram

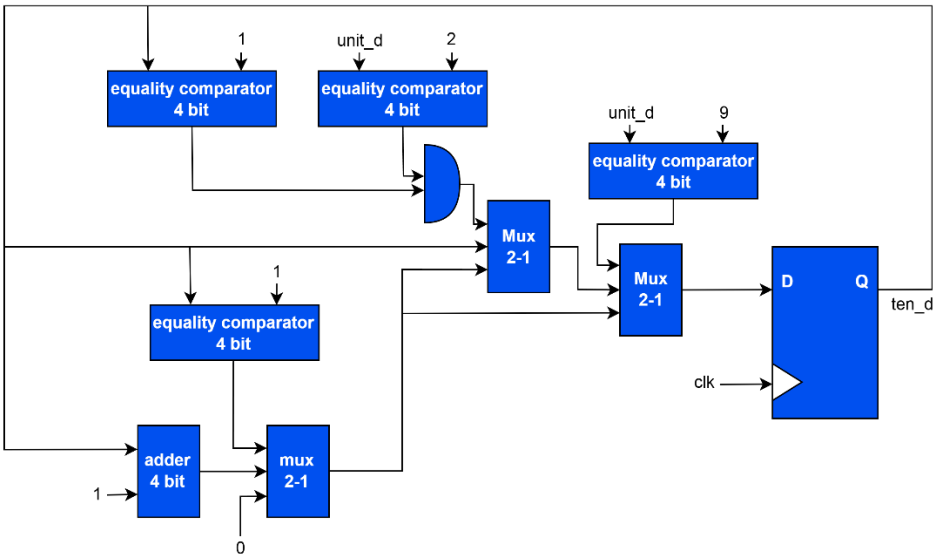


Figure 33. month_counter block diagram

This block is responsible for counting months and depends on the day counter. When the day counter reaches 31 for months 1, 3, 5, 7, 8, 10, and 12; or 30 for months 4, 6, 9, and 11; or 28 for February in a non-leap year and 29 in a leap year, the month counter increases by one. It continues until it reaches 12 months.

3.2.5.1 Interface signal

Signal Name	Width	I/O	Description
clk	1	Input	System clock signal
reset_n	1	Input	System asynchronous reset, active LOW
carry_d	1	Input	Signal to indicate minute is day to day_in_month
inc	1	Input	Signal to increase time
dec	1	Input	Signal to decrease time
select_time	1	Input	Signal to chosse time to edit
ten_m	4	Output	Ten day counter
unit_m	4	Output	Unit day counter
carry_m	1	Output	Signal to indicate month is count to 12

Table 11. month_counter signals description

3.2.6. year_counter

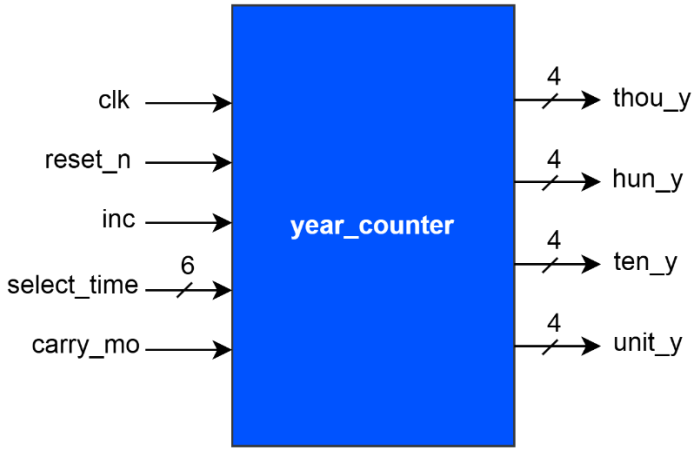


Figure 34. year_counter block diagram

This block is responsible for counting years and depends on the month counter. When the month counter reaches 12 months, the year counter increases by one.

3.2.5.1 Interface signal

Signal Name	Width	I/O	Description
clk	1	Input	System clock signal
reset_n	1	Input	System asynchronous reset, active LOW
carry_d	1	Input	Signal to indicate minute is day to day_in_month
inc	1	Input	Signal to increase time
dec	1	Input	Signal to decrease time
select_time	1	Input	Signal to chosse time to edit
thou_y	4	Output	Thou year counter
hun_y	4	Output	Hun year counter
ten_y	4	Output	Ten year counter
unit_y	4	Output	Unit year counter

Table 12. day_counter signals description

3.2.7. day_in_month_decode

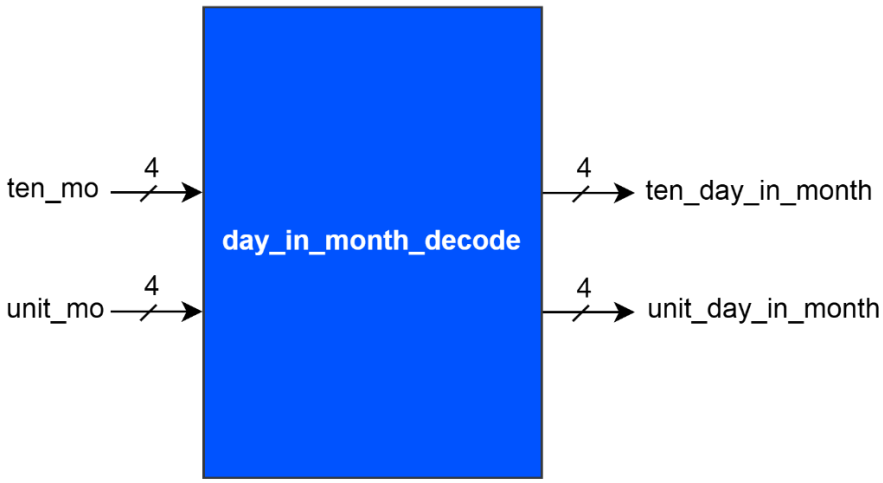


Figure 35. day_in_month_decode block diagram

3.2.7.1. Interface signal

Signal Name	Width	I/O	Description
ten_mo	4	Input	Ten month counter
unit_mo	4	Input	Unit month counter
ten_day_in_month	4	Output	Ten day in month counter
unit_day_in_month	4	Output	Unit day in month counter

Table 13. leap_year_check signals description

3.2.8. leap_year_check

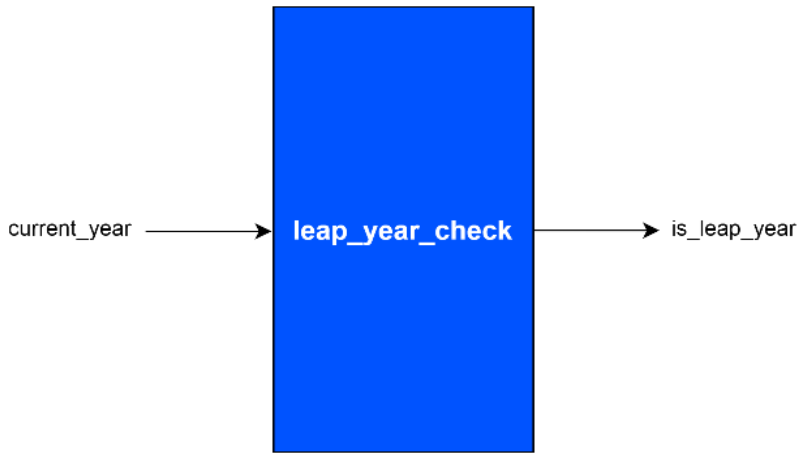


Figure 36. leap_year_check block diagram

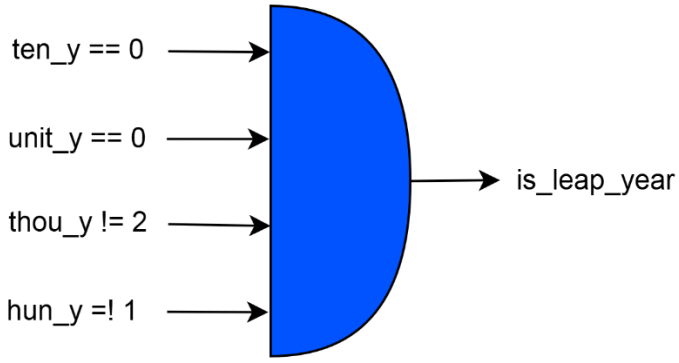


Figure 37. leap_year_check block diagram

3.2.8.1. Interface signal

Signal Name	Width	I/O	Description
thou_y	4	Input	Thou year counter
hun_y	4	Input	Hun year counter
ten_y	4	Input	Ten year counter
unit_y	4	Input	Unit year counter
is_leap_year	1	Output	Signal indicate year is leap

Table 14. leap_year_check signals description

3.3. top_display

The top_display block shows time and date on the seven-segment LEDs. It receives 4-bit inputs for seconds, minutes, hours, day, month, and year, then decodes and displays them on the corresponding LED digits.

The block is controlled by mode_display and mode_alarm signals:
+ Mode_display selects which information (time, date, or year) is shown.
+ Mode_alarm activates the alarm display or alarm output signal.

Each LED group represents a specific part of the clock, ensuring clear and accurate display according to the selected mode.

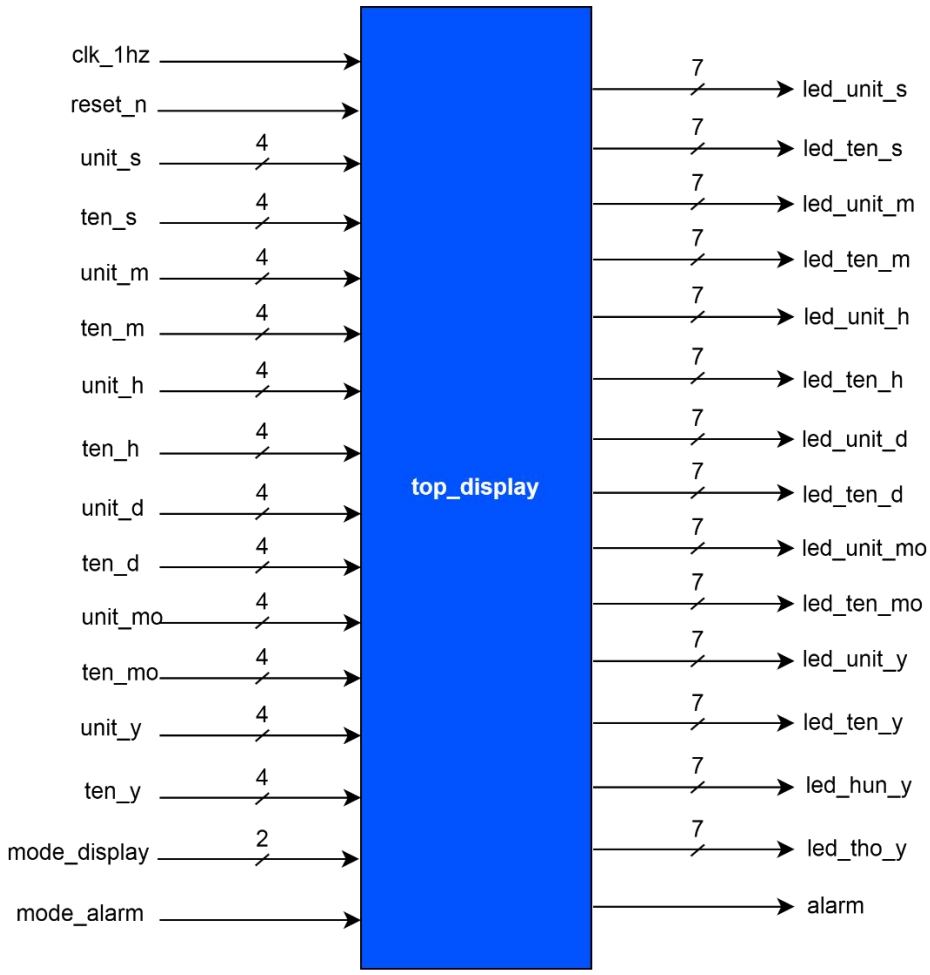


Figure 38. top_display block diagram

3.3.1 seven_seg_decoder

This block is responsible for decoding each digit into the proper signals for the seven-segment display. It converts the values of seconds, minutes, and hours into the corresponding segments so that the correct numbers appear on the display. This allows users to easily read the current time from the clock.

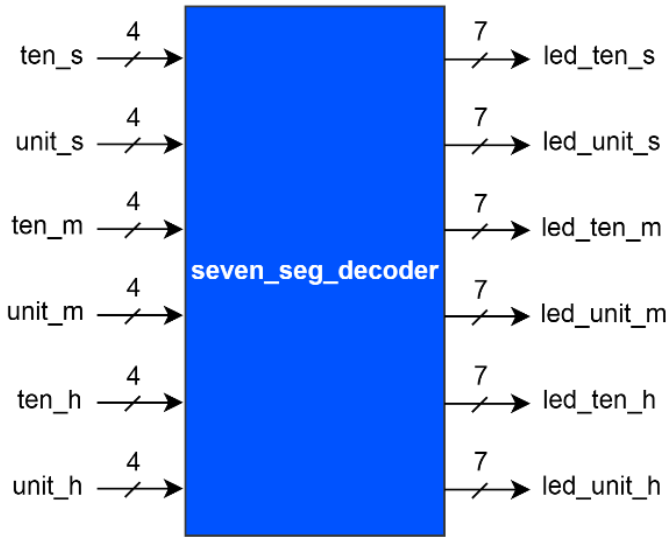


Figure 39. seven_seg_decoder block diagram

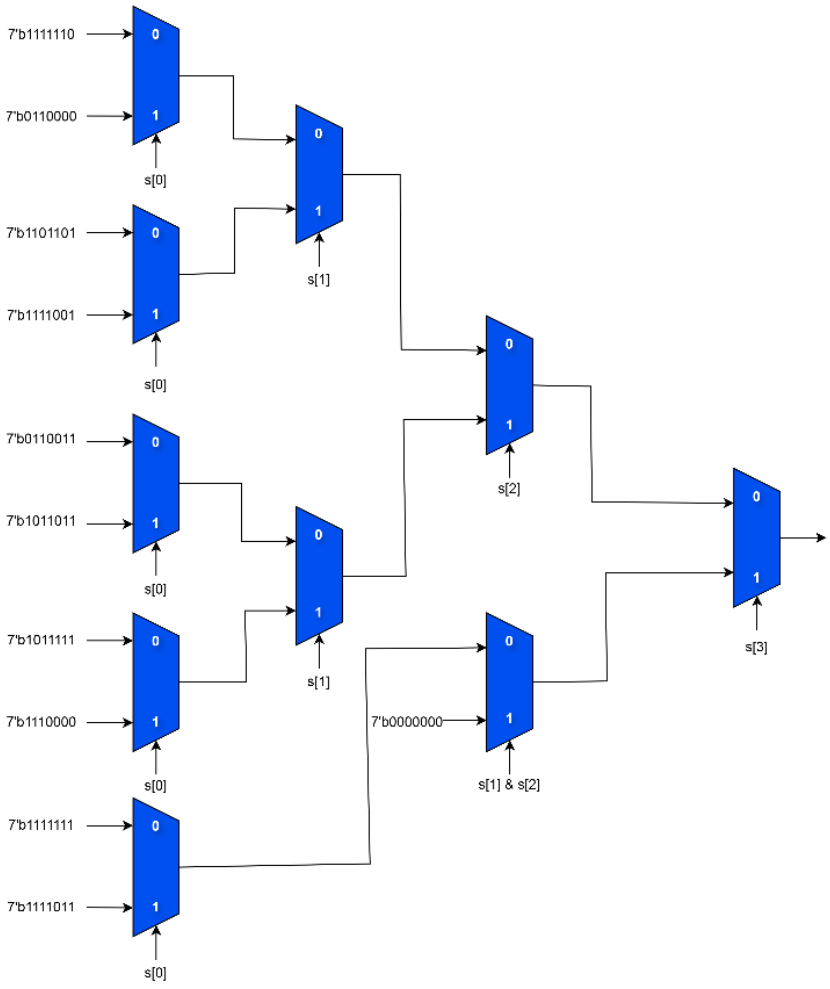


Figure 40. seven_seg_decoder block diagram

3.3.1.1 Interface signal

Signal Name	Width	I/O	Description
unit_s	4	Input	Second units
ten_s	4	Input	Second tens
unit_m	4	Input	Minute units
ten_m	4	Input	Minute tens
unit_h	4	Input	Hour units
ten_h	4	Input	Hour tens
led_unit_s	7	Output	Seven seg second tens
led_ten_s	7	Output	Seven seg second units
led_unit_m	7	Output	Seven seg minute tens
led_ten_m	7	Output	Seven seg minute units
led_unit_h	7	Output	Seven seg hour tens
led_ten_h	7	Output	Seven seg hour units
led_unit_y	7	Output	Seven seg year units
led_ten_y	7	Output	Seven seg year tens
led_hun_y	7	Output	Seven seg yearhunds
led_thou_y	7	Output	Seven seg year thous

Table 15. seven_seg_decoder signals description

4. Timing diagram

4.1. count_second

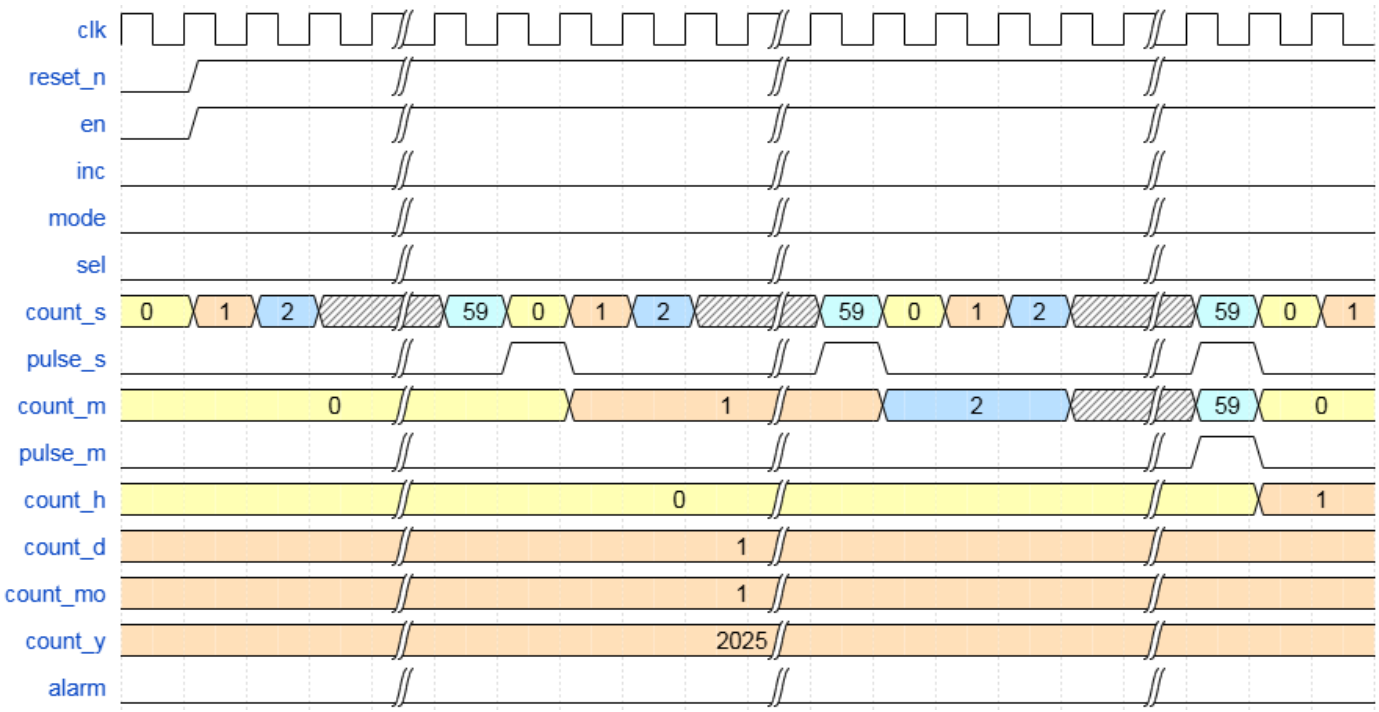


Figure 41. timing diagram

The timing diagram illustrates how the second, minute, and hour counters increase over time and how they depend on one another. When the second counter reaches 60, it resets to zero and increases the minute counter by one. Similarly, when the minute counter reaches 60, it resets to zero and increases the hour counter by one. This diagram helps visualize the relationship and synchronization between the different time units in the clock system.